

technologies for legitimate enterprise management purposes. Intel's ME was designed specifically as "a business-class feature intended for businesses to manage their fleets of computers". Large organisations need efficient ways to manage hundreds or thousands of computers remotely, including the ability to troubleshoot problems, install updates, and maintain security policies without requiring physical access to each machine.

Intel's Active Management Technology, which runs on the ME, enables IT administrators to perform management tasks even when computers are powered off or have corrupted operating systems. This capability proves invaluable for businesses managing large computer deployments, as it reduces the need to physically visit each problematic machine.

AMD's PSP works differently, focusing on security over remote management. The PSP handles security functions, including boot processes, encryption key management, and firmware validation. AMD's approach centres on creating a secure foundation for the system, ensuring that malicious software cannot compromise fundamental system operations.

The PSP manages several security features, including Secure Memory Encryption (SME) and Secure Encrypted Virtualization (SEV). These let the processor encrypt system memory automatically, protecting data even if an attacker gains physical access to the computer's RAM.

Technical capabilities that raise eyebrows

The capabilities of these embedded processors have understandably caused concern among privacy advocates and security researchers. Intel's ME possesses remarkably broad system access that extends far beyond typical hardware components.

According to security expert Damien Zammit's research, Intel's ME maintains full access to system memory without the main CPU's

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Embargoed until Feb. 27, 2024 at 6 pm CET. 1. As of February 2024, based on security by design principles, enabled handling of vulnerabilities, robust Intel Platform Update process, an extensive bug bounty program, as well as internal research through Red Team and Purple Team. Learn more at intel.com/security

FIRST TIME ON PERFORMANCE A-SERIES: AMD SECURE PROCESSOR
DEDICATED SECURITY SUBSYSTEM INTEGRATED WITHIN AMD 6TH GENERATION A-SERIES PROCESSORS

PSP HARDWARE INCLUDES:

- ▲ Dedicated 32-bit microcontroller (ARM® Cortex®-a5)
- ▲ Isolated on-chip ROM & SRAM
- ▲ Access to system memory / resources
- ▲ OTP for platform-unique key material
- ▲ HW logic for secure control of x86 core boot
- ▲ Cryptographic co-processor
 - RSA (up to 16384-bit)
 - Sha (sha1, sha-224, sha-256, sha-512)
 - ECC (basic mathematical computations, up to 384-bit)
 - AES engine (ECB, CBC, CFB, OFB, CTR, GCM, CMAC, GMAC, IAPM, XTS-AES128)
 - Zlib (decompression)
 - TRNG (basis for RDRAND)

The diagram shows the PSP hardware architecture with components: 32-bit Micro, MMU, I/D Cache, Local Memory Interface, ROM, SRAM, INT, Crypto (RSA, AES, SHA, ECC), and DMA.

27 | AMD TECH DAY 2023 | Embargo lifts June 2nd at 9:00 PM Eastern U.S. time

knowledge or consent. This means the ME can read, write, or modify any data in your computer's RAM whilst your main processor remains completely unaware of these activities.

The ME's network capabilities are equally concerning. It maintains full access to the TCP/IP stack and can send and receive network packets independently of your operating system's firewall protection. Even if you've configured sophisticated firewall rules to block unauthorised network traffic, the ME operates completely outside these restrictions. It maintains a dedicated connection to the network interface and can communicate over the internet regardless of your system's security settings.

Intel's ME remains active even when your computer appears completely shut down. As long as the system receives standby power from being

plugged in, the ME continues running its own firmware. This persistent operation occurs independently of your computer's power state, meaning the ME could theoretically monitor or communicate even when you believe your system is completely offline.

AMD's PSP operates quite differently, with more limited but still significant capabilities. The PSP runs independently from the main CPU using its own dedicated ROM and SRAM memory. However, AMD's PSP lacks network access capabilities. This architectural decision significantly reduces the potential for remote exploitation, as any attacks would require physical access to the target system.

Examining the backdoor claims

The debate over whether these technologies constitute backdoors has